AARAV RATHI

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EDUCATION

University of California, Berkeley Expected May 2029

B.A. Data Science, College of Computing, Data Science, and Society

Relevant Coursework: CS 61A, CS 61B, CS 70, Data 100, Math 54

Fremont High School — Sunnyvale, CA 2025

EXPERIENCE

ContextQA — *Applied AI Engineer Intern* June – August 2026

- Built testlens, a Neo4j knowledge graph mapping GitHub PRs and Jira/Linear tickets to affected test cases.
- Replaced full-suite regression runs with targeted test selection, cutting tests executed per change by 83%.
- Exposed the graph via an OAuth-secured MCP server, letting the internal LLM agent query change impact.

Aisera — *Engineering Intern* June – July 2024

- Automated LLM testing workflows using Python, reducing manual QA effort across the quality engineering team.
- Improved indexing and retrieval performance in the retrieval-augmented generation (RAG) system.

PROJECTS

SML Training Pipeline — *Applied AI take-home project* github.com/Aarav360/sml-training-pipeline

- Fine-tuned Qwen3 0.6B for multi-step tool use on a 12GB GPU with QLoRA (4-bit, rank 16); SFT on 5K filtered xLAM examples raised tool-call format accuracy from 19% to 100%.
- Designed a GRPO environment with an oracle-obfuscated reward (tool outputs transformed by a quadratic mod a large prime), making reward hacking through internal computation infeasible at 0.6B.
- Diagnosed flat GRPO reward (std = 0 across both runs) as single-pass generation in TRL's GRPOTrainer never returning tool results to context; specified the multi-turn rollout fix.

Digital Math Teacher — *Next.js*, *TypeScript*, *FastAPI*, *PostgreSQL* github.com/Aarav360/Digital-Math-Teacher

- Built a full-stack math practice workspace: Next.js 15 / TypeScript frontend on a FastAPI and PostgreSQL backend with async SQLAlchemy, Alembic migrations, and JWT auth (guest tokens and Google OAuth).
- Implemented a custom canvas engine — freehand and shape tools, lasso and box selection, undo/redo across strokes, text and images, cursor-anchored zoom — with debounced autosave and a localStorage fallback on server failure.

RESEARCH

Independent Research — advised by Emily Sheetz (PhD candidate, University of Michigan) Aug 2023 – Jan 2024

- Authored "Exploring N-Gram Models for Adaptive Predictive Texts"; accepted to the Synopsys Science Fair Championship (Mar 2024).

Research Assistant — Utkarsh Contractor (VP of AI, Aisera; Senior Research Fellow, Stanford) Aug – Dec 2022

- Supported "Deep Multimodal Learning for Surveillance," working with the CLIP open-source neural network.
- Developed an object classification pipeline over public video footage, evaluating detection and false-positive rates.

LEADERSHIP & SERVICE

Founder, DEPO Mar 2024 – Present

- Built and ran computer science education programming for the Prajna Foundation, growing its outreach and funding.
- Led discussion groups with underserved teenagers in India on depolarization, bias reduction, and social listening.

Team Operations & Equipment Manager, Berkeley Badminton May 2026 – Present

- Tested shuttlecock candidates for speed and durability, cutting season supply cost 40%, from $50 to $30 per tube.
- Ran setup and operations for Golden Bear and Polar Bear, Berkeley's two annual tournaments (1,000+ participants).

ADDITIONAL

Programming: Python, Java, SQL, JavaScript

Frameworks & Tools: PyTorch, Hugging Face, scikit-learn, pandas/NumPy, Neo4j/Cypher, FastAPI, React, Node.js, Git

Languages: English, Hindi, Spanish

Athletics: Badminton — 15+ tournaments with multiple medal finishes; HS Varsity Captain (2024, 2025)